

Exhaust & Ventilation Spec Sheet

Doc No: HVAC-005

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Project: Lakefront Office Tower — Phase II

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SECTION 23 34 23 — HVAC POWER VENTILATORS

1.0 SCOPE OF WORK

Furnish and install all exhaust fans, relief fans, roof ventilators, and associated roof curbs, gravity back-draft dampers, motorized isolation dampers, and controls per drawings M4.01–M4.03. Coordinate all roof penetrations with the roofing contractor (Section 07 54 00).

2.0 EQUIPMENT SCHEDULE

Tag	Service	CFM	SP (in.)	Motor (HP)	Drive	Discharge
EF-1	Toilet Exhaust Floors 3–6	4,200	0.50	3	Direct	Vertical cap
EF-2	Toilet Exhaust Floors 7–10	4,200	0.50	3	Direct	Vertical cap
EF-3	Toilet Exhaust Floors 11–14	4,200	0.50	3	Direct	Vertical cap
EF-4	Toilet Exhaust Floors 15–18	3,600	0.50	3	Direct	Vertical cap
EF-5	Parking Garage Exhaust	28,000	1.00	20	VFD	Horizontal louver
EF-6	Kitchen Hood Exhaust	6,500	1.50	10	Direct	Vertical cap
EF-7	Electrical Room Exhaust	2,400	0.50	2	Direct	Vertical cap
EF-8	Janitor / Storage Exhaust	800	0.25	0.5	Direct	Vertical cap
RF-1	Building Relief — West	18,000	0.75	15	VFD	Horizontal louver
RF-2	Building Relief — East	18,000	0.75	15	VFD	Horizontal louver

3.0 SYSTEM SCHEMATIC / DRAWING

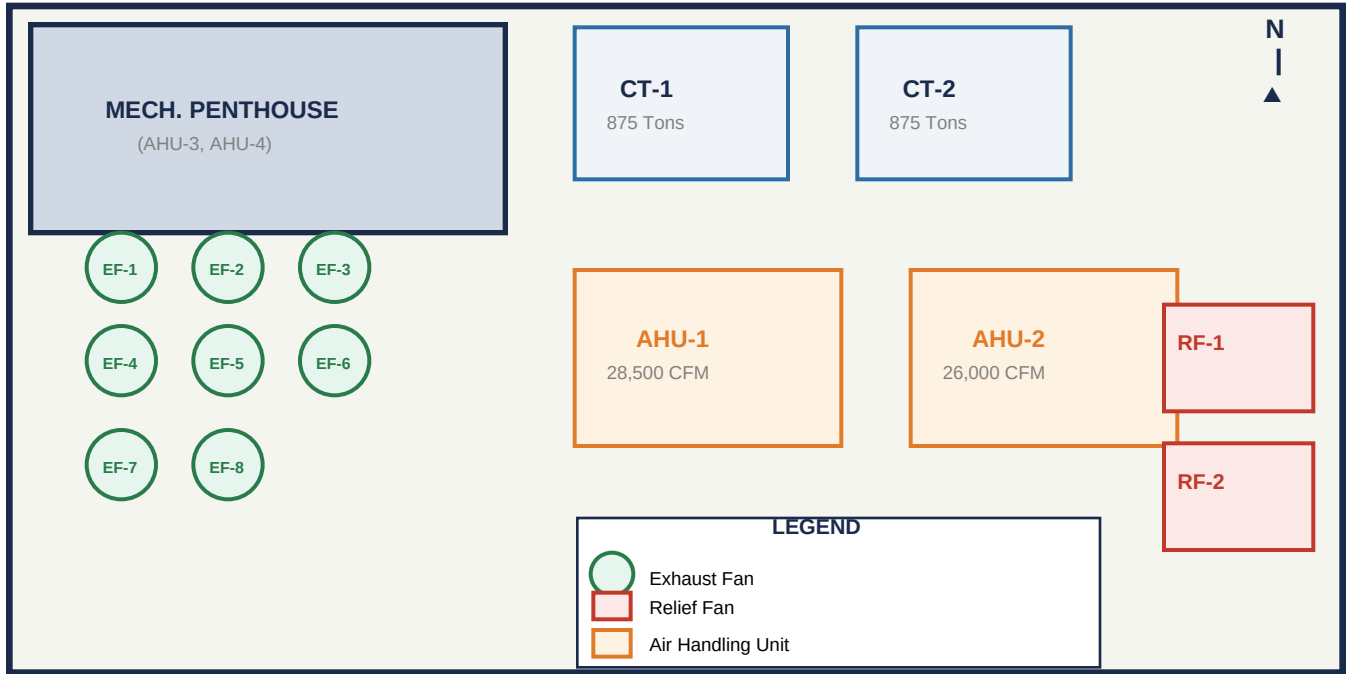


Figure 5 — Rooftop Mechanical Equipment Plan (Schematic — Not to Scale)

4.0 TECHNICAL SPECIFICATIONS

4.1 Exhaust Fan Construction

Performance	AMCA 210 licensed for air, AMCA 300 sound-rated
Housing	Galvanized steel, continuously welded, AMCA Arrangement 4
Wheel	Backward-inclined centrifugal; aluminum or galvanized steel
Motors	NEMA Premium Efficiency, inverter-duty rated where VFD driven
EF-6 (Kitchen)	UL 762-listed, grease-tight construction per NFPA 96
EF-5 (Parking)	CO/NO _x sensor modulation per IBC §404.1 and IMC 404

4.2 Roof Curbs & Dampers

Roof Curbs	Factory-fabricated, 18-in. minimum height, insulated, cant strips
Back-Draft Dampers	Gravity, low-leakage, 0.10 cfm/sf at 1-in. w.g.
Isolation Dampers	Motorized, parallel blade, 24V actuator, spring-return closed
Vibration Isolation	Neoprene-in-shear pads at all curb mounting flanges

5.0 INSTALLATION NOTES

- All fans: AMCA 210 licensed for air performance and AMCA 300 sound-rated.
- Vibration isolation per ASHRAE HVAC Applications Handbook Chapter 48.
- All roof penetrations: factory-fabricated curbs, 18-in. minimum height.
- EF-6 (Kitchen): grease-tight construction, access panels every 12 ft per NFPA 96.
- EF-5 (Parking): CO/NO_x sensors at 3 levels; modulate fan speed to maintain < 25 ppm CO.
- Balance all exhaust systems per ASHRAE 111; TAB report required before occupancy.
- RF-1 and RF-2: modulate VFD to maintain building static pressure at +0.05 in. w.g.

